

Operational Research II, Midterm 2, 2026 Spring

- Duration: 120 minutes
- Closed material, No calculator

- Name: _____
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- Write legibly.
- Justification is necessary unless stated otherwise.

1	20
2	20
3	15
4	15
Total	70

#1. Fill in the blanks by using the following terms. [Each 2pts]

periodic	aperiodic	class	recurrent	transient	absorbing
finite	infinite	irreducible	reducible	reach	communicate

(a)	
(b)	
(c)	
(d)	
(e)	
(f)	
(g)	
(h)	
(i)	
(j)	

- An absorbing state is a special case of a [(a)] state.
- A DTMC is said to be [(b)] if there exists a state whose period is higher than one.
- A DTMC is said to be [(c)] if there are more than one class.
- If all diagonal elements of transition matrix in DTMC are positive, then the DTMC has to be [(d)].
- A DTMC is said to be [(e)] if all states communicate.
- A group of states that can reach each other is said to be a [(f)].
- In a DTMC, the limiting probability is same as the unique stationary distribution regardless of initial states if following three conditions of the DTMC are met: [(g)] [(h)] [(i)].
- State i is said to be [(j)] if, starting from state i , the probability of getting back to state i in some future is equal to 1.

#2. A small bank is staffed by a single server. It has been observed that, during a normal business day, the inter-arrival times of customers to the bank are iid having exponential distribution with mean 3 minutes. Also, the processing times of customers are iid having the following distribution (in minutes):

$$P(X = 1) = \frac{1}{4}, \quad P(X = 2) = \frac{1}{2}, \quad P(X = 3) = \frac{1}{4}.$$

An arrival finding the server busy joins the queue. The waiting space is infinite.

(a) What is long-run average waiting time of each customer in the queue?[10pts]

(b) What is long-run average number of customers in the bank?[10pts]

#3. This problem concerns whether the following statement is true or false.

If a DTMC is aperiodic and irreducible, then there exists a stationary distribution.

(a) Circle one of the following: [5pts] (**True/False**)

(b) If you answered "true", then explain why it is true and provide an example. If you answered "false", then provide a counterexample and explain it. [10pts]

#4. Consider a DTMC with the following transition matrix.

$$\mathbf{P} = \begin{pmatrix} .25 & .5 & & .25 & & & \\ .5 & & .5 & & & & \\ & .5 & .25 & & .25 & & \\ & & & 1 & & & \\ & & .5 & .5 & & & \\ & & & & & .3 & .7 \\ & & & & .7 & & .3 \\ & & & & .3 & .7 & \end{pmatrix}$$

(a) How many classes does it have? [5pts]

(b) Calculate $\mathbf{P}^{100}$ (Show your work with proper justification.) [10pts]

